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CS402 MCQs lecture 35 to 45

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1) L= language of words containing even number of a's. Regular Expression is

(a+b)*aa(a+b)*
(b+ab*a)*
a+bb*aab*a
(a+b)*ab(a+b)*

2) The regular expression defining the language L1 ∪ L2 can be obtained, converting and reducing the previous ---------- into a ---------- as after eliminating states.

GTG, TG
FA, GTG
FA, TG
TG, RE

3) The language that can be expressed by any regular expression is called a Non regular language.
True **False**

4) The languages ---------- are the examples of non regular languages.

PALINDROME and PRIME
PALINDROME and EVEN-EVEN
EVEN-EVEN and PRIME
FACTORIAL and SQURE

5) Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form [IMG]file:///C:/DOCUME%7E1/ADMINI%7E1/LOCALS%7E1/Temp/msohtml1/01/clip_image009.gif[/IMG]for n=1,2,3, … are the words in L. called.
Complement of L
Pumping Lemma
Kleene's theorem
None in given

(6) Languages are proved to be regular or non regular using pumping lemma.
True False

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(7) ---------- is obviously infinite language.
EQUAL-EQUAL
EVEN-EVEN
PALINDROME
FACTORIAL

(8) **If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.**
True False

(9) **Myhill Nerode theorem is consisting of the followings,**

L partitions Σ* into distinct classes.
If L is regular then, L generates finite number of classes.
If L generates finite number of classes then L is regular.
All of above

(10) **The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.**
R=Q/P **Q=R/P** Q=P/R P=R/Q

(11) If two languages R and Q are given, then **the prefixes of Q in R** denoted by **Pref(Q in R).**
True False

(12) **Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to,**

{b,bbba,bbbaaa}
{b,bba,bbaaa}
{ab,bba,bbbaa}
{b,bba,bbba}

(13) **If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.**

Non-regular
Equal
Regular
Infinite

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(14) **"CFG" stands for _____**

Context Free Graph
Context Free Grammar
Context Finite Graph
Context Finite Grammar

(15) _____ states are called the halt states.

ACCEPT and REJECT
ACCEPT and READ
ACCEPT AND START
ACCEPT AND WRITE

(16) The part of an FA, where the input string is placed before it is run, is called _____

State
Transition
Input Tape
Output Tape

(17) In new format of an FA (discussed in lecture 37), This state is like dead-end non final state

ACCEPT
REJECT
STATR
READ

(18) For language L defined over {a, b}, then L partitions {a, b}* into ..... classes

Infinite
Finite
Distinct
Non-distinct

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(19) The major problem in the earliest computers was

To store the contents in the registers
To display mathematical formulae
To load the contents from the registers
To calculate the mathematical formula

(20) Between the two consecutive joints on a path

One character can be pushed and one character can be popped
Any no. of characters can be pushed and one character can be popped
One character can be pushed and any no. of characters can be popped
Any no. of characters can be pushed and any no. of characters can be popped

(21) In pumping lemma theorem (x y^n z) the range of n is

n=1, 2, 3, 4…….
n=0, 1, 2, 3, 4…….
n=…….-3,-2,-1, 0, 1, 2, 3, 4……
n=…….-3,-2,-1, 1, 2, 3, 4……

(22) The PDA is called non-deterministic PDA when there are more than one out going edges from…… state

START or READ
POP or REJECT
READ or POP
PUSH or POP

(23) Identify the TRUE statement:

A PDA is non-deterministic, if there are more than one READ states in PDA
A PDA is never non-deterministic
Like TG, A PDA can also be non-deterministic
A PDA is non-deterministic, if there are more than one REJECT states in PDA

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(24) There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite.

True False

(25) If an effectively solvable problem has answered in yes or no, then this solution is called ---------

Decision procedure
Decision method
Decision problem
Decision making

(26) The following problem(s) ---------- is/are called decidable problem(s).

The two regular expressions define the same language
The two FAs are equivalent
Both a and b
None of given

(27) To examine whether a certain FA accepts any words, it is required to seek the paths from ------- state.

Final to initial
Final to final
Initial to final
Initial to initial

(28) The high level language is converted into assembly language codes by a program called compiler.
TRUE FALSE

(29) Grammatical rules which involve the meaning of words are called ----------
Semantics
Syntactic
Both a and b
None of given
(30) Grammatical rules which do not involve the meaning of words are called ----------
Semantics
Syntactic
Both a and b
None of given

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(31) The symbols that can't be replaced by anything are called ----------
Productions
Terminals
Non-terminals
All of above

(32) The symbols that must be replaced by other things are called _____
Productions
Terminals
Non-terminals
None of given

(33) The grammatical rules are often called_____
Productions
Terminals
Non-terminals
None of given

(34) The terminals are designated by _____ letters, while the non-terminals are designated by _____ letters.
Capital, bold
Small, capital
Capital, small
Small, bold

(35) The language generated by _____ is called Context Free Language (CFL).

FA TG **CFG** TGT

36) **Σ = {a,b} Productions S→XaaX X→aX X→bX X→Λ**
This grammar defines the language expressed by_____
(a+b)*aa(a+b)*
(a+b)*a(a+b)*a
(a+b)*aa(a+b)*aa
(a+b)*aba+b)*

(37) S → aXb|b XaX → aX|bX|Λ The given CFG generates the language in English _____
Beginning and ending in different letters
Beginning and ending in same letter
Having even-even language
None of given

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(38)**The CFG is not said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,**

TRUE **FALSE**

(39) The language generated by that CFG is regular if _____
No terminal → semi word
No terminal → word
Both a and b
None of given

(40) The production of the form no terminal → Λ is said to be null production.
TRUE FALSE

(41) A production is called null able production if it is of the form N → Λ
TRUE FALSE

(42) The productions of the form nonterminal → one nonterminal, is called _____
Null production
Unit production
Null able production
None of given

(43) CNF is stands for
Context Normal Form
Complete Normal Form
Chomsky Normal Form
Compared Null Form

The End

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